

# Assignment III

## (Laplace Transform)

**Subject:** Calculus-II (MTH 210)

**Deadline: 2080-09-22**

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1. Find the Laplace transform of the following functions:

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|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| (i) $\sin 2t \cos 3t$                                                                                                               | <b>Ans:</b> $\frac{2(s^2 - 5)}{(s^2 + 1)(s^2 + 25)}$                                           |
| (ii) $\sinh at \cos bt$                                                                                                             | <b>Ans:</b> $\frac{1}{2} \left[ \frac{s-a}{(s-a)^2 + b^2} - \frac{s+a}{(s+a)^2 + b^2} \right]$ |
| (iii) $t^2 \cos \omega t$                                                                                                           | <b>Ans:</b> $\frac{2s(\omega^2 - 3s^2)}{(s^2 + \omega^2)^3}$                                   |
| (iv) $te^{-3t} \cos 2t$                                                                                                             | <b>Ans:</b> $\frac{s^2 + 6s + 5}{(s^2 + 6s + 13)^2}$                                           |
| (v) $\frac{\sinh t}{t}$                                                                                                             | <b>Ans:</b> $\frac{1}{2} \log\left(\frac{s+1}{s-1}\right)$                                     |
| (vi) $\frac{e^{-at} - e^{-bt}}{t}$                                                                                                  | <b>Ans:</b> $\log\left(\frac{s+b}{s+a}\right)$                                                 |
| (vii) $\sin 2t u_\pi(t)$                                                                                                            | <b>Ans:</b> $-\frac{2e^{-\pi s}}{s^2 + 4}$                                                     |
| (viii) $(t-1)^2 u(t-1)$                                                                                                             | <b>Ans:</b> $\frac{2e^{-s}}{s^3}$                                                              |
| (ix) $f(t) = \begin{cases} 1 & \text{if } 0 < t < \pi \\ 0 & \text{if } \pi < t < 2\pi \\ \sin t & \text{if } t > 2\pi \end{cases}$ | <b>Ans:</b> $\frac{1 - e^{-\pi s}}{s} + \frac{e^{-2\pi s}}{s^2 + 1}$                           |

2. Find the inverse Laplace transform of:

- |                                                                 |                                                         |
|-----------------------------------------------------------------|---------------------------------------------------------|
| (i) $\cot^{-1}\left(\frac{s}{\omega}\right)$                    | <b>Ans:</b> $\frac{\sin \omega t}{t}$                   |
| (ii) $\log\left(\frac{s(s+1)}{s^2+4}\right)$                    | <b>Ans:</b> $\frac{2 \cos 2t - 1 - e^t}{t}$             |
| (iii) $\log\left(\frac{s+a}{s-a}\right)$                        | <b>Ans:</b>                                             |
| (iv) $\frac{1}{s^2(s^2 + \omega^2)}$ (by using convolution)     | <b>Ans:</b> $\frac{\omega t - \sin \omega t}{\omega^3}$ |
| (v) $\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$ (by using convolution) | <b>Ans:</b> $\frac{a \sin at - b \sin bt}{a^2 - b^2}$   |

3. Solve the following initial value problem by using Laplace transform:

- |                                                                 |                                                  |
|-----------------------------------------------------------------|--------------------------------------------------|
| (i) $9y'' - 6y' + y = 0 \quad y(0) = 3, y'(0) = 1$              | <b>Ans:</b> $y = 3e^{\frac{t}{3}}$               |
| (ii) $x'' + 2x' + 5x = e^{-t} \sin t \quad x(0) = 0, x'(0) = 1$ | <b>Ans:</b> $\frac{e^{-t}(\sin t + \sin 2t)}{3}$ |

$$(iii) \quad y'' - 2y' + y = e^t \quad y(0) = 2, y'(0) = -1 \quad \textbf{Ans: } e^t(2 - 3t + \frac{t^2}{2})$$

$$(iv) \quad x'' + x = t \cos 2t \quad x(0) = x'(0) = 0 \quad \textbf{Ans: } \frac{1}{9}(4 \sin 2t - 5 \sin t - 3t \cos 2t)$$

$$(v) \quad y'' + 3y' + 2y = 2u(t - 2) \quad y(0) = y'(0) = 0 \quad \textbf{Ans:}$$

**Hints:**  $\mathcal{L}[2u(t - 2)] = \frac{2e^{-2s}}{s}$